

ACG-WAM: World-Action Modeling via Action-Conditioned Geometric Latent Prediction

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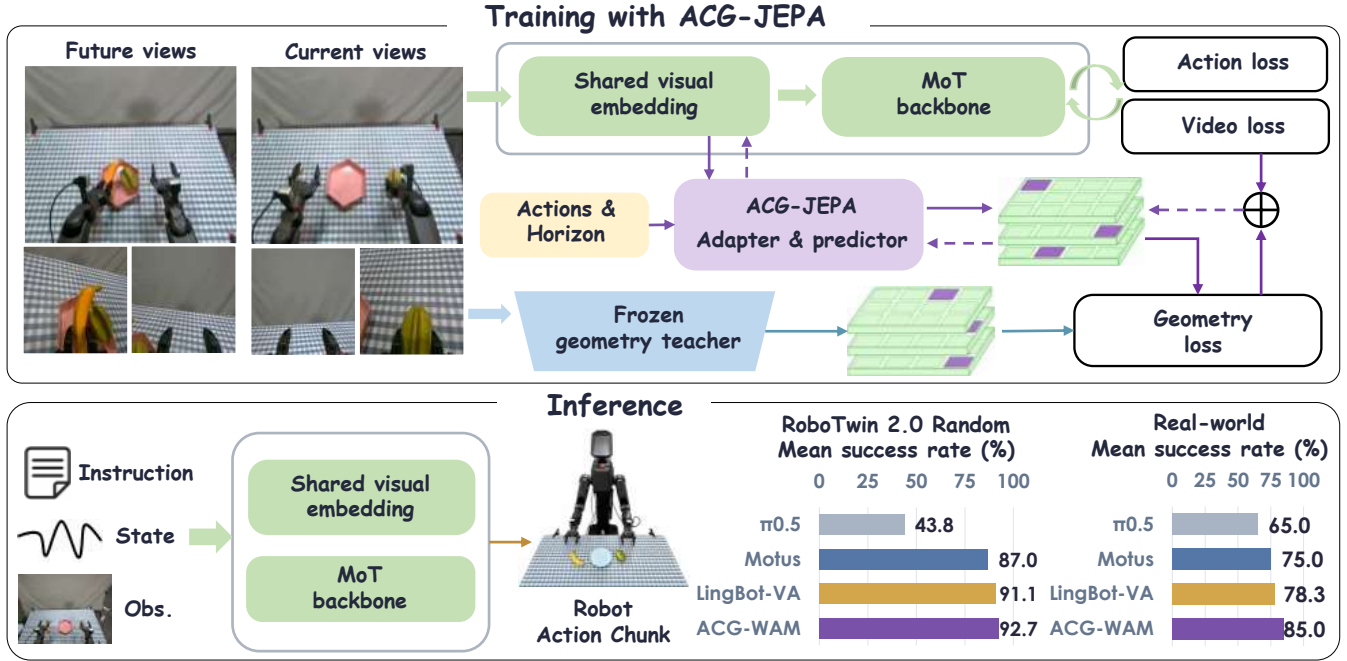


Fig. 1. Overview of ACG-WAM. Visual Geometry Grounded Transformer (VGGT) supplies geometric targets for prediction from current visual features and actions. The auxiliary loss updates the shared visual embedding during training; inference uses the trained backbone.

Abstract—World action models jointly learn visual prediction and robot actions, providing a way to use observations of scene evolution for policy learning. Their video and action losses, however, provide no explicit target for the geometric consequences of a demonstrated action sequence. Moreover, visual features taken after temporal attention can contain future observations, making them unsuitable as the sole current visual input to an auxiliary predictor. We introduce ACG-WAM and its auxiliary objective, the Action-Conditioned Geometric Joint-Embedding Predictive Architecture (ACG-JEPA), which predicts geometric features at several horizons from the current observation and intervening actions, using the future slot of a frozen VGGT encoding of each current and future image pair as the target. We apply this supervision from the head and wrist cameras to a shared visual embedding before temporal mixing, and remove the teacher and auxiliary modules at inference. On 50 RoboTwin 2.0 tasks, ACG-WAM achieves 93.46% success in clean scenes, with the best randomized success (92.68%) and mean across both settings (93.07%) among the compared methods; across three tasks on a real robot, it achieves 85.00% success and 91.67% partial completion score, exceeding Motus by 10.00 and 9.17 percentage points, respectively.

I. INTRODUCTION

Robot manipulation requires actions that change the scene in a useful way: a grasp must secure an object, and a placement must leave it in the intended position. Gen-

erative policies learn action sequences from images and language instructions [1]–[4]. World action models (WAMs) also predict future observations, coupling visual forecasting with action generation [5]–[7]. This joint formulation gives policy training access to how objects and robot hands move between observations, in addition to the demonstrated control commands. For manipulation, the relevant change is often a spatial relation: a hand closes around an object, a container receives it, or one object becomes supported by another. Spatial representations have proved useful for such tasks [8]–[10]. In bimanual manipulation, these relations also connect the two arms: moving an object with one arm changes the conditions for the other’s next action. The question is how to make geometric information a useful training signal for a WAM while retaining its existing observation and action interface.

Two challenges arise when adding this supervision. First, video and action losses do not explicitly require a representation of the geometric change associated with an action interval. From the same observation, a robot can approach, lift, or move an object; the starting image alone does not specify which outcome a predictor should learn. An additional prediction target must therefore describe a future

scene in the context of both its starting observation and the actions that connect them. Even over the same duration, different action sequences can lead to different stages of an interaction. The action interval identifies the demonstrated movement, while its duration specifies how far ahead the predictor should look. Second, the location of the supervised policy feature matters. During joint video and action training, attention mixes tokens from current and future frames. A token assigned to the current frame can consequently contain future visual information after this mixing. Using that token as the visual input to an auxiliary predictor would let it use part of the observation it is supposed to predict. For this supervision to benefit action generation, its gradients must reach the visual features shared with the policy.

These challenges motivate two complementary choices. For the first, we predict a geometric representation of a future endpoint conditioned on the intervening action sequence. A frozen teacher jointly encodes the two images, allowing the future representation to use the current scene as context. Sampling different endpoints associates the same current observation with both short interactions and longer action sequences. This exposes the visual embedding to changes that develop over different portions of a manipulation sequence, from the initial approach to later object transport. For the second, we supervise the visual embedding before the backbone mixes frames. The student then receives features computed from the current image alone, while the teacher can use the future image to form a target. The head camera describes the shared workspace, including the target region and the relation between the arms; wrist cameras reveal local configurations of the grippers and manipulated objects. Together, these views provide spatial evidence about both the overall arrangement and the immediate interaction. Supervising the shared embedding with these views connects geometric prediction to the visual input used by the deployed policy.

We implement these choices as ACG-WAM, a Motus policy trained with the ACG-JEPA objective. Frozen VGGT [11] encodes the current and future images jointly within each camera. We retain its future slot, pool the features from each view, and train a predictor using the current visual features, demonstrated actions, and a temporal horizon. The auxiliary loss updates the shared visual embedding alongside the base video and action objectives. The teacher and predictor are discarded at inference, so the policy uses its original inputs and action generation path. Figure 1 shows this connection between training and deployment. We evaluate the resulting policy on 50 RoboTwin 2.0 tasks and three tasks on a real robot. It reaches 93.46% clean success and leads the compared methods in randomized success (92.68%) and the mean across both settings (93.07%), with 85.00% mean success on the robot. Ablations on six simulation tasks examine action conditioning, temporal horizons, and the geometric target. Together, these evaluations test whether geometric endpoint supervision benefits task completion under visual variation and execution on physical hardware.

In summary, our main contributions are:

- Geometric dynamics conditioned on actions. We introduce a joint embedding objective that predicts geometric targets over several horizons from the current observation and intervening actions. A frozen teacher jointly encodes current and future images to provide explicit supervision for world action modeling.
- Geometric distillation from multiple views. We connect targets from the head and wrist cameras to policy features computed from the current observation before temporal mixing. Auxiliary gradients update the shared visual embedding, while the teacher and predictor are removed at deployment.
- Benchmark and real robot evaluation. We evaluate the policy on 50 RoboTwin 2.0 tasks and three manipulation tasks on a real robot. It achieves the best randomized and mean success among the compared methods on RoboTwin 2.0 and the highest average SR and PCS on the robot. Ablations on six simulation tasks support the Joint target design, action conditioning, and supervision at multiple horizons.

II. RELATED WORK

A. Generative Policies and World Action Models

Robot policies combine pretrained representations of images and language with action learning. RT-2 and OpenVLA use different approaches to transfer [2], [3]; π_0 and $\pi_{0.5}$ focus on continuous control and adaptation [4], [12]. UniPi uses video generation for planning, and GR-1 predicts video and actions [13], [14]. Unified World Models couple video and action diffusion [7]. Motus combines understanding, generation, and control in a latent WAM, while LingBot-VA studies causal world modeling for robot control [5], [6]. We use Motus as the backbone and add geometric supervision during training.

B. Predictive Representations and Alignment

Joint embedding methods predict features instead of reconstructing pixels. I-JEPA and V-JEPA study prediction in learned image and video feature spaces [15], [16]; V-JEPA 2 extends this approach to world modeling and planning [17]. DINO-WM learns dynamics from pretrained visual features [18], and JEPA-WAM predicts a joint representation of current and future observations in a pretrained V-JEPA space [19]. REPA aligns diffusion representations with features from an external encoder [20]. We use joint prediction to supervise geometric features from VGGT. The predictor connects a demonstrated action interval to its geometric endpoint, while supervision before temporal attention directly trains the shared embedding of the current observation. This connects latent prediction to the visual representation used for action generation.

C. Geometric Representations for Manipulation

DUST3R and MAST3R recover spatial structure and correspondences; VGGT predicts several geometric quantities with a feed-forward transformer [11], [21], [22]. RVT, 3D

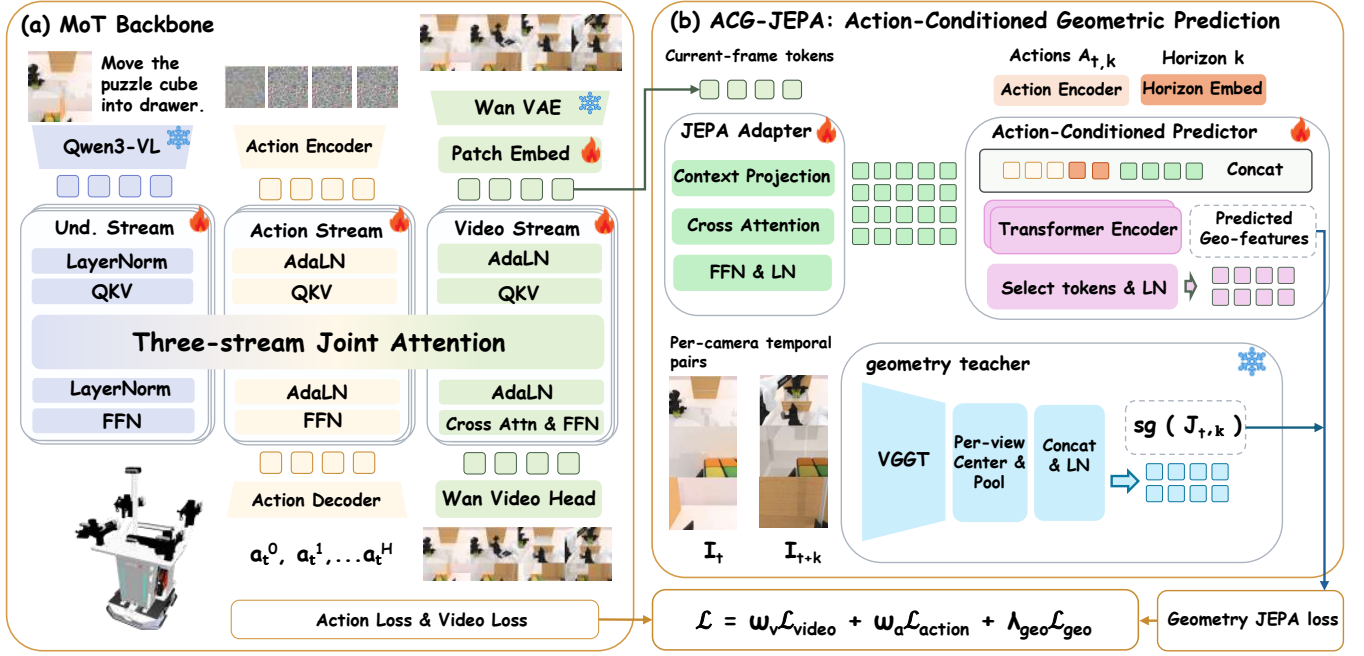


Fig. 2. Training architecture of ACG-WAM. Motus predicts video and actions. The auxiliary predictor uses features of the current frame, demonstrated actions, and a horizon to predict the geometric target. Future images provide teacher targets without entering the student’s visual input. The auxiliary loss updates the shared patch embedding; the base losses train the backbone.

Diffusion Policy (DP3), and 3D Diffuser Actor use spatial representations for control [8]–[10]. WAM4D introduces spatial registers to predict future depth during training, and MECo-WAM uses a frozen VGGT teacher, action-aware relational distillation, and restricted expert visibility [23], [24]. Both remove their auxiliary geometry components at inference. We formulate supervision as geometric endpoint prediction from the current observation, an intervening action interval, and its horizon. Targets jointly encode current and future images within each camera, and supervision at multiple horizons trains the shared visual embedding used by the deployed policy before temporal attention.

III. METHOD

A. Overview

ACG-WAM augments Motus with an auxiliary task that predicts geometric features from the current observation and a sequence of actions. At time t , the policy receives camera observations $\{I_t^v\}_{v \in \mathcal{V}}$, robot state s_t , and a language instruction ℓ , and generates a chunk of H actions. The Motus backbone [5] uses Mixture-of-Transformers (MoT) attention to couple visual understanding, latent video generation [25], and action generation. Video and action flow matching [26] define the base loss:

$$\mathcal{L}_{\text{base}} = w_v \mathcal{L}_{\text{video}} + w_a \mathcal{L}_{\text{action}}. \quad (1)$$

For the auxiliary task, let $J_{t,k}$ denote a frozen geometric representation computed jointly from the current and future observations across the cameras. Let Z_t denote the policy feature computed from the current observation. The predictor learns

$$(Z_t, \mathbf{A}_{t,k}, k) \mapsto J_{t,k}, \quad (2)$$

Here t and k index the downsampled video sequence. Let r be the number of sampled actions per video transition and $\mathbf{a}_t^{(i)}$ the i th action in the chunk starting at I_t , indexed from zero. Then $\mathbf{A}_{t,k} = (\mathbf{a}_t^{(0)}, \dots, \mathbf{a}_t^{(kr-1)})$ contains exactly kr actions aligned with I_{t+k} . With raw-frame start f_t and action sampling stride s , action i is sampled at $f_t + (i+1)s$, and the target image at $f_t + krs$. This alignment supplies the predictor with the action prefix for each selected geometric endpoint. Sec. III-B defines the prediction objective, and Sec. III-C describes the teacher targets and student features.

B. Geometric Prediction Conditioned on Actions

We form the target by feeding the current image and a future image to a frozen geometric teacher in a single forward pass. Its attention allows the representation of the future image to incorporate information from the current scene. We retain the features at the future slot, giving the representation $J_{t,k}$ defined in Sec. III-C. The target is

$$D_{t,k} = \text{sg}(J_{t,k}), \quad (3)$$

where sg stops gradients through the teacher. Within this auxiliary task, future images are used to construct the target; the student’s visual input is computed from the current observation alone.

Selecting the future slot gives the predictor a geometric endpoint represented in the context of the initial scene. Joint teacher attention allows features at that endpoint to incorporate both observations, so the supervision connects the starting configuration to the demonstrated outcome in a shared latent representation.

The predictor also receives the actions connecting the observations. From the same scene, the robot may approach,

lift, or move an object, depending on the action sequence. We combine an adapted visual feature $A_\eta(Z_t)$, an action embedding $E_a(\mathbf{A}_{t,k})$, and a horizon embedding e_k :

$$Q_{t,k} = g_\theta(A_\eta(Z_t), E_a(\mathbf{A}_{t,k}), e_k). \quad (4)$$

The action encoder flattens the sequence in time-major order, adaptively average-pools the resulting vector to a fixed width, and applies two linear layers with a GELU activation and layer normalization to obtain one token. The fixed temporal layout gives each pooling bin a consistent position within the action sequence. A transformer processes this action token, a learned horizon token, and queries derived from the current visual features, combining the action interval with the observed scene. Its normalized query outputs form $Q_{t,k}$. The original policy action branch continues to learn from the language instruction and the base loss.

During training, one horizon is sampled uniformly for each batch and shared by its examples. Each future endpoint is paired with the demonstrated action prefix spanning the same interval. The horizon embedding distinguishes short interactions from the consequences of longer action sequences. All horizons use the same predictor, target dimensions, and loss.

For $Q = Q_{t,k}$ and $D = D_{t,k} \in \mathbb{R}^{M \times d}$, let $\bar{Q}$ and $\bar{D}$ be their means over the M target slots. We use

$$\begin{aligned} \mathcal{L}_{\text{geo}} = & \alpha[1 - \cos_\epsilon(\bar{Q}, \bar{D})] + \beta \frac{\|Q - D\|_F^2}{Md} \\ & + \frac{\gamma}{M} \sum_{j=1}^M [1 - \cos_\epsilon(Q_j, D_j)], \end{aligned} \quad (5)$$

where $\cos_\epsilon$ is stabilized cosine similarity, and α , β , and γ weight the three terms.

The loss supervises the representation at two levels: pooled cosine similarity aligns the overall feature direction, while squared error and slotwise cosine similarity match individual feature slots. Squared error penalizes coordinate differences after normalization, and slotwise cosine similarity aligns the direction of each predicted slot with its teacher target. Teacher normalization uses fixed scale and bias; the predictor’s output normalization learns these parameters per channel.

C. Geometric Supervision of the Current Observation

The teacher provides targets from three cameras, and the student predictor reads the shared visual embedding before MoT attention. This placement lets the geometric loss update a feature extractor used by the deployed policy.

a) Targets from multiple cameras: The head camera captures scene layout and the relation between the arms; the wrist cameras capture local interactions with objects. We process the head, left wrist, and right wrist views separately, $\mathcal{V} = \{h, l, r\}$. For each camera, frozen VGGT [11] receives the current image followed by the future image:

$$(X_{t|t,k}^v, X_{t+k|t,k}^v) = \Psi([I_t^v, I_{t+k}^v]). \quad (6)$$

We use the deepest available VGGT aggregator output and select its future slot $X_{t+k|t,k}^v$, which can attend to both

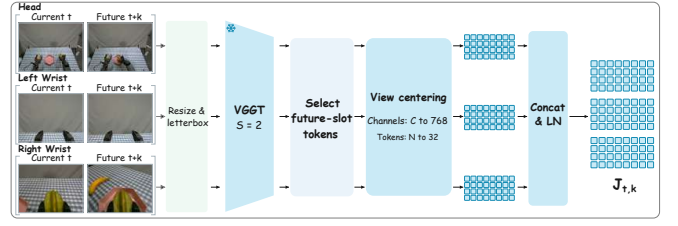


Fig. 3. Construction of the geometric target with frozen VGGT. Each camera contributes an ordered pair of current and future images. The future slot is centered and pooled within each view before the three views are concatenated and normalized. The target branch is used only during training.

images. We abbreviate it as $X_{t,k}^v$. Each channel is centered across the view’s N tokens:

$$\tilde{X}_{t,k,n}^v = X_{t,k,n}^v - \frac{1}{N} \sum_{j=1}^N X_{t,k,j}^v. \quad (7)$$

Adaptive average pooling P first reduces the channel axis to $d = 768$ and then the token axis to 32 slots per camera, without learned parameters. We concatenate the views in head, left wrist, and right wrist order and apply layer normalization independently over each slot’s feature axis, without learned scale or bias:

$$J_{t,k} = \text{LN}([P(\tilde{X}_{t,k}^h); P(\tilde{X}_{t,k}^l); P(\tilde{X}_{t,k}^r)]) \in \mathbb{R}^{M \times d}. \quad (8)$$

Here M is the total number of pooled slots and d is their feature width. The teacher mixes temporal information within each camera pair. Separate slots retain the identity of each view after concatenation. No subtraction of independently encoded endpoints is used in this target.

Figure 3 details this construction. In our implementation, pooling produces 32 slots of width 768 for each camera, giving $M = 96$ after concatenation. Centering removes the mean across tokens within each view, and pooling produces the fixed number of target slots.

b) Features of the current observation: Future video latents participate in the backbone’s denoising loss. After attention mixes frames and streams, even tokens assigned to the current frame can contain future visual information. We therefore read the student feature before the first MoT layer.

The frozen video VAE, $\mathcal{E}$, encodes the current image separately. Its latent serves as the clean conditioning frame. The shared patch embedding F_ϕ , applied to each frame, produces

$$Z_t = F_\phi(\mathcal{E}(I_t)). \quad (9)$$

In the joint computation, Z_t is the slice corresponding to the current frame. Separate encoding and a patch embedding with no temporal mixing ensure that it depends on I_t alone. The auxiliary predictor receives this feature, the demonstrated actions, and the horizon. Future video latents remain inputs to the base training objective.

The adapter A_η pools and projects the input channels, adds learned spatial embeddings, and resamples the patches using cross-attention from learned queries. A residual feed-forward block and normalization complete the adapter. Its fixed set

TABLE I

SUCCESS RATES (%) ON ROBOTWIN 2.0. BOLD MARKS THE BEST RESULT IN EACH COLUMN OF (A) AND EACH METHOD PAIR IN (B). Δ IS IN PERCENTAGE POINTS.

(a) Overall performance on 50 tasks				(b) Selected task improvements over Motus						
Method	RoboTwin 2.0			Task	Clean			Random		
	Clean	Random	Mean		Motus [5]	ACG-WAM	Δ	Motus [5]	ACG-WAM	Δ
GO-1 [27]	37.80	36.24	37.02	Hanging Mug	38	70	+32	38	76	+38
$\pi_{0.5}$ [12]	42.98	43.84	43.41							
X-VLA [28]	72.80	72.84	72.82	Handover Mic	78	100	+22	63	98	+35
Motus [5]	88.66	87.02	87.84	Move Can Pot	34	98	+64	74	90	+16
LingBot-VA [6]	91.99	91.11	91.55	Place Mouse Pad	66	94	+28	68	81	+13
MECo-WAM [24]	93.26	91.98	92.62	Scan Object	67	92	+25	66	87	+21
WAM4D [23]	93.82	89.86	91.84	Open Microwave	95	98	+3	91	99	+8
ACG-WAM (ours)	93.46	92.68	93.07							

of output queries predicts the teacher slots while attending to spatially indexed features of the current frame.

c) *Training and inference*: The total loss is

$$\mathcal{L} = \mathcal{L}_{\text{base}} + \lambda_{\text{geo}} \mathcal{L}_{\text{geo}}. \quad (10)$$

The geometric loss updates the adapter, action encoder, predictor, and shared patch embedding F_ϕ . The embedding also supplies the video stream, so it receives gradients from both $\mathcal{L}_{\text{base}}$ and $\mathcal{L}_{\text{geo}}$. The teacher and VAE remain frozen. Downstream MoT layers and the action decoder are optimized by the base loss using the updated embedding. This shared parameter path carries the auxiliary supervision into the policy.

At inference, the policy uses the trained Motus backbone. The teacher, target cache, adapter, and predictor are removed; the shared visual embedding remains. Geometric supervision therefore adds no module to the deployed computation graph.

IV. EXPERIMENTS

We evaluate ACG-WAM on RoboTwin 2.0 and on three tasks with a real robot. The main comparisons measure task success, and the ablations compare prediction targets, action conditioning, and temporal horizons.

A. Implementation

a) *Model implementation*: The backbone uses Wan2.2-TI2V-5B and Qwen3-VL-2B-Instruct [29], [30]. Frozen VGGT targets are cached offline following Sec. III-C, resizing wrist crops to the head-view size before preprocessing. The 96-query adapter and predictor have one and two blocks, respectively, with width 768, eight heads, feed-forward width 3072, and zero dropout. The action encoder has two 768 $\times$ 768 linear layers. Geometric losses use float32.

b) *Training*: We initialize ACG-WAM from the pre-trained base model and train on RoboTwin for 40k updates with ACG-JEPA enabled throughout. This matches the budget reported in Motus [5]; LingBot-VA uses 50k updates [6]. We use joint future-slot targets, $k \in \{1, 2, 4, 8\}$, $r = 2$, and $s = 3$: the eight-transition window contains $H = 16$ actions,

with six raw frames per video transition. AdamW optimizes Eq. (10) with base learning rate 5×10^{-5} , weight decay 0.01, and moments (0.9, 0.95). After 200 warmup updates, the learning-rate multiplier decays linearly from 0.99 to 0.4. Four GPUs with microbatch four and 16-step gradient accumulation give a global batch of 256. We use gradient clipping 0.5 and bfloat16 mixed precision. Loss weights are $\lambda_{\text{geo}} = 0.01$ and $(\alpha, \beta, \gamma) = (1, 0.05, 1)$, with unit weights for the base losses.

c) *RoboTwin inference*: We use ten UniPC steps with shift 1.0 for video and action generation, executing all 16 joint-position actions before the next policy observation.

B. Simulation Experiments

a) *Setting*: We evaluate on RoboTwin 2.0 [31], which comprises 50 bimanual manipulation tasks. The *demo-clean* and *demo-randomized* settings test nominal scenes and visually randomized environments, respectively. The task suite spans object relocation, stacking, tool interaction, and coordinated bimanual manipulation.

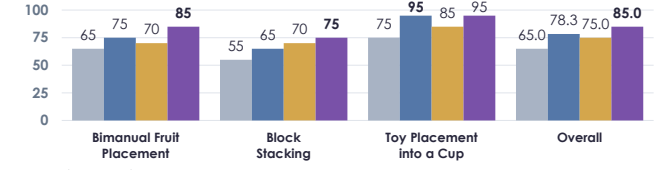


Fig. 4. Successful ACG-WAM trajectories for Hanging Mug (top), Place A to B (Right) (middle), and Place Bread in Skillet (bottom) in randomized RoboTwin 2.0 scenes. Red boxes mark the manipulated objects in the first frame of each sequence.

b) *Baselines*: Table I(a) compares ACG-WAM with GO-1, $\pi_{0.5}$, X-VLA, Motus, LingBot-VA, MECo-WAM, and WAM4D. Results for the first four methods are taken from

Real-world evaluation

(a) Success rate (%)



(b) Partial completion score (%)

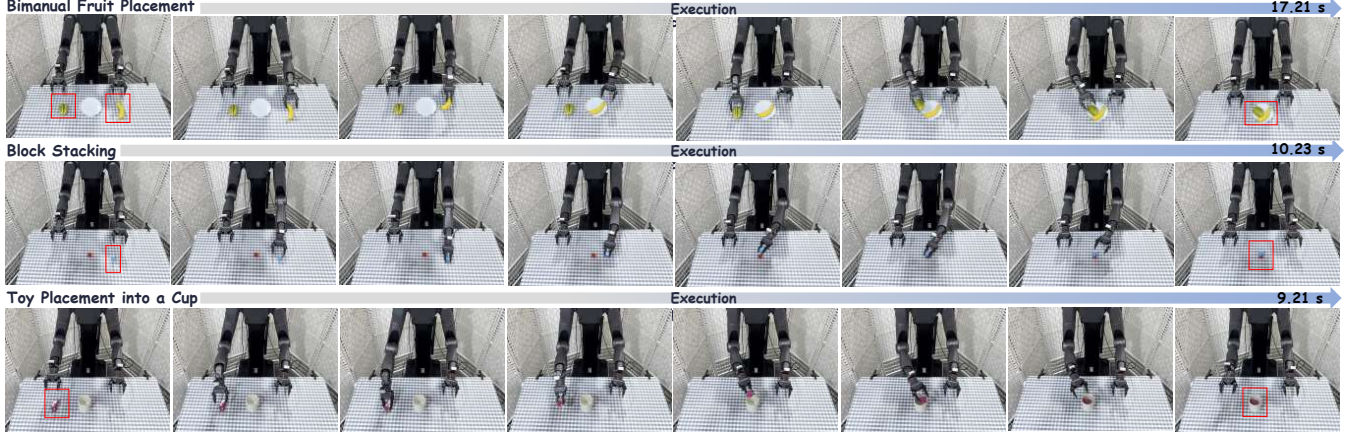
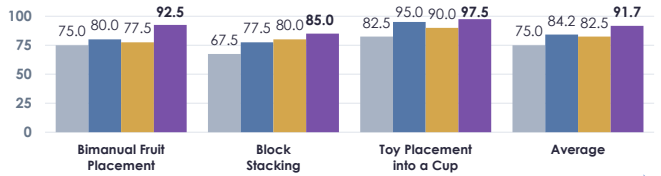


Fig. 5. Evaluation on a real robot across three manipulation tasks. The upper panels show success rate and partial completion score. Below, eight keyframes per task show execution from left to right; arrows indicate elapsed time.

Motus [5]; the remaining baseline results are from their respective publications [6], [23], [24].

c) *Evaluation*: For ACG-WAM, each task is evaluated in each setting over 100 episodes with unseen instruction templates and evaluation seed 42. Success follows the benchmark’s completion criterion for each task. Table I reports macro averages over all 50 tasks for each setting and six task comparisons with Motus. Mean gives equal weight to Clean and Random.

d) *Results*: ACG-WAM achieves 93.46% clean success and the highest randomized success (92.68%) and mean across both settings (93.07%) among the compared methods in Table I(a). It exceeds MECo-WAM in both settings and achieves clean performance close to WAM4D, with a 2.82-point advantage under randomization. Relative to Motus, geometric supervision improves clean and randomized success by 4.80 and 5.66 percentage points.

Across all 50 tasks, the clean and randomized win/tie/loss counts relative to Motus are 31/7/12 and 37/5/8, respectively. Table I(b) illustrates these gains across distinct interactions, including hanging, handover, relocation, scanning, and opening an articulated object. The randomized gains reach 38 points on *Hanging Mug* and 35 points on *Handover Mic*. The ablations in Sec. IV-D compare geometric targets, action conditioning, and temporal horizons.

e) *Qualitative examples*: Figure 4 shows successful randomized episodes of *Hanging Mug*, *Place A to B (Right)*, and *Place Bread in Skillet*. The sequences illustrate completion of three spatial goals: supporting a mug on a rack, positioning one object relative to another, and placing bread within a receptacle.

C. Experiments on a Real Robot

a) *Platform and tasks*: We evaluate bimanual manipulation on the TRON2 platform equipped with WUJI hands. A head camera observes the workspace, and two wrist cameras capture local interactions with objects. The task suite consists of *Bimanual Fruit Placement*, *Block Stacking*, and *Toy Placement into a Cup*. Together, these tasks cover coordinated object placement, stable stacking, and placement within a confined receptacle. Figure 5 shows the evaluation overview and example task executions. We compare ACG-WAM with $\pi_{0.5}$ [12], LingBot-VA [6], and Motus [5] under a common protocol for demonstrations on this platform, observation preprocessing, and action representation. We collect 100 demonstration episodes per task and fine-tune the policies for 15k updates on the corresponding task data.

b) *Evaluation*: Each policy is evaluated over 20 trials per task, giving 60 trials per policy. We report task success rate (SR) and a normalized partial completion score (PCS) to distinguish complete execution from partial progress. Each trial receives a score $s_{j,i} \in \{0, 1, 2\}$: zero for no completion, one for partial completion, and two for full completion. For N_j trials on task j ,

$$\text{SR}_j = \frac{100}{N_j} \sum_i \mathbf{1}[s_{j,i} = 2], \quad \text{PCS}_j = \frac{100}{2N_j} \sum_i s_{j,i}. \quad (11)$$

For *Bimanual Fruit Placement*, the scores count zero, one, or two fruits stably placed in the target area. For *Block Stacking*, partial credit denotes placing the upper block before it falls; full credit requires the released stack to remain stable for at least three seconds. For *Toy Placement into a Cup*, partial credit denotes lifting the toy, while full credit requires

TABLE II
SUCCESS RATE (SR) AND PARTIAL COMPLETION SCORE (PCS) ON THE REAL ROBOT (%). BEST RESULTS IN EACH COLUMN ARE BOLD.

Model	Bimanual Fruit Placement		Block Stacking		Toy Placement into a Cup		Task Average	
	SR	PCS	SR	PCS	SR	PCS	SR	PCS
$\pi_{0.5}$ [12]	65	75.0	55	67.5	75	82.5	65.00	75.00
LingBot-VA [6]	75	80.0	65	77.5	95	95.0	78.33	84.17
Motus [5]	70	77.5	70	80.0	85	90.0	75.00	82.50
ACG-WAM (ours)	85	92.5	75	85.0	95	97.5	85.00	91.67

releasing it into the cup and retaining it there for at least three seconds. The reported average gives equal weight to the three tasks. Table II pairs SR and PCS for the compared policies.

c) *Results*: ACG-WAM achieves the highest average SR and PCS among the evaluated policies, reaching 85.00% and 91.67%, respectively (Table II). These exceed Motus by 10.00 and 9.17 percentage points, extending the simulation gains to physical bimanual manipulation. It leads both metrics on *Bimanual Fruit Placement* and *Block Stacking*. On *Toy Placement into a Cup*, it ties LingBot-VA at 95% SR and achieves the highest PCS of 97.5%. Together, the SR and PCS gains show improved complete execution and progress through the prescribed manipulation stages.

D. Ablation Studies

We examine three design choices in ACG-JEPA: the geometric prediction target, action conditioning, and the temporal horizons used for supervision. All variants are trained for 8k updates on the same six tasks: *Handover Mic*, *Hanging Mug*, *Move Can Pot*, *Place Mouse Pad*, *Put Object Cabinet*, and *Scan Object*. Each task is evaluated over 100 episodes per setting with seed 42. Tables III and IV report average success rates. The Mean column reports the average of the clean and randomized success rates. All comparisons share the same full Joint checkpoint.

a) *Effect of prediction horizons*: We compare sampling from $k \in \{1, 2, 4, 8\}$ with a fixed short horizon ($k = 1$) and a fixed long horizon ($k = 8$). All three variants retain the joint target and action conditioning. Here k counts downsampled video transitions within the training window; the action prefix is aligned with the selected target endpoint.

TABLE III
PREDICTION HORIZON ABLATION ON ROBOTWIN 2.0. VALUES ARE SUCCESS RATES (%).

Horizon set	Clean	Random	Mean
Short {1}	79.33	79.17	79.25
Long {8}	82.17	80.67	81.42
Multiple {1, 2, 4, 8}	84.67	82.67	83.67

Sampling multiple horizons gives the highest mean success, 83.67%, in Table III. It exceeds the fixed long and short horizons by 2.25 and 4.42 percentage points, respectively. The difference is particularly clear on *Hanging Mug*:

sampling multiple horizons reaches 46% clean and 38% randomized success, compared with 39%/28% for the long horizon and 22%/20% for the short horizon. Supervision at multiple horizons links each current observation to action intervals of different lengths, covering both short interactions and longer manipulation sequences.

b) *Effect of the prediction target*: We compare the final Joint target with subtraction of endpoint features. Both variants use action conditioning, sample the same horizons, and read student features before temporal mixing. Joint uses the teacher’s future slot in Eq. (3). Residual encodes each endpoint independently with the same frozen VGGT and the same pooling for each view, then forms

$$D_{t,k}^{\text{res}} = \text{sg}(U_{t+k} - U_t), \quad (12)$$

where U_s pools and normalizes the features from all camera views at time s . The predictor output shape and loss family are unchanged.

TABLE IV
TARGET AND ACTION CONDITIONING ABLATIONS ON ROBOTWIN 2.0. VALUES ARE SUCCESS RATES (%).

Variant	Clean	Random	Mean
Endpoint residual	79.83	81.00	80.42
Joint, no actions	83.00	80.83	81.92
Joint (full)	84.67	82.67	83.67

Using the jointly encoded future target increases mean success from 80.42% to 83.67%, a gain of 3.25 percentage points. The clean and randomized gains are 4.83 and 1.67 points, respectively. Joint improves four tasks in each setting; on *Hanging Mug*, success rises from 30%/32% to 46%/38%, while on *Place Mouse Pad* it rises from 82%/80% to 89%/84%. These results show the advantage of the Joint target design over endpoint subtraction for geometric supervision. Joint teacher attention constructs the future representation with access to the initial scene, providing an endpoint target that carries temporal context.

c) *Effect of action conditioning*: We replace the geometric predictor’s action sequence with zeros before the action encoder, retaining the Joint target, horizon sampling, model architecture, and policy loss. In Table IV, action conditioning increases mean success from 81.92% to 83.67%,

a gain of 1.75 percentage points, or 21 additional successes over 1,200 episodes. Clean and randomized success improve by 1.67 and 1.83 points, respectively. The largest gain occurs on *Hanging Mug*, where randomized success rises from 27% to 38%. These results support conditioning geometric prediction on the intervening action sequence: actions specify how the current scene evolves, making geometric supervision more effective for manipulation.

V. CONCLUSION

We presented ACG-WAM to give world action models explicit supervision for the geometric consequences of robot actions. Its ACG-JEPA objective predicts jointly encoded geometric endpoints from the current observation and intervening actions at several horizons. Supervision from the head and wrist cameras updates the policy’s shared visual embedding before temporal mixing. This supervision lets the visual embedding learn from the geometry of completed transitions while remaining computable from the robot’s current observations.

Evaluations on RoboTwin 2.0 and three real robot tasks demonstrate the value of this supervision for manipulation. ACG-WAM leads the compared methods in randomized and mean benchmark success and achieves the highest average SR and PCS on the real robot. The ablations support the Joint target design, action conditioning, and prediction at several horizons, showing how these choices contribute to the final policy’s success. Together, these results establish geometric endpoint prediction as an effective way to connect world modeling with action generation. The resulting policy benefits from geometric supervision while retaining the backbone’s observation interface and computation at deployment.

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REFERENCES

- [1] C. Chi *et al.*, “Diffusion policy: Visuomotor policy learning via action diffusion,” in *Proc. Robotics: Science and Systems (RSS)*, 2023.
- [2] B. Zitkovich *et al.*, “RT-2: Vision-language-action models transfer web knowledge to robotic control,” in *Proc. Conf. Robot Learn. (CoRL)*, ser. Proc. Mach. Learn. Res., vol. 229, 2023, pp. 2165–2183.
- [3] M. J. Kim *et al.*, “OpenVLA: An open-source vision-language-action model,” in *Proc. Conf. Robot Learn. (CoRL)*, ser. Proc. Mach. Learn. Res., vol. 270, 2025, pp. 2679–2713.
- [4] K. Black *et al.*, “ π_0 : A vision-language-action flow model for general robot control,” in *Proc. Robotics: Science and Systems (RSS)*, 2025.
- [5] H. Bi *et al.*, “Motus: A unified latent action world model,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 35 101–35 113.
- [6] L. Li *et al.*, “Causal world modeling for robot control,” in *Proc. Robotics: Science and Systems (RSS)*, 2026.
- [7] C. Zhu, R. Yu, S. Feng, B. Burchfiel, P. Shah, and A. Gupta, “Unified world models: Coupling video and action diffusion for pretraining on large robotic datasets,” in *Proc. Robotics: Science and Systems (RSS)*, 2025.
- [8] Y. Ze, G. Zhang, K. Zhang, C. Hu, M. Wang, and H. Xu, “3D diffusion policy: Generalizable visuomotor policy learning via simple 3D representations,” in *Proc. Robotics: Science and Systems (RSS)*, 2024.
- [9] A. Goyal, J. Xu, Y. Guo, V. Blukis, Y.-W. Chao, and D. Fox, “RVT: Robotic view transformer for 3D object manipulation,” in *Proc. Conf. Robot Learn. (CoRL)*, ser. Proc. Mach. Learn. Res., vol. 229, 2023, pp. 694–710.
- [10] T.-W. Ke, N. Gkanatsios, and K. Fragkiadaki, “3D diffuser actor: Policy diffusion with 3D scene representations,” in *Proc. Conf. Robot Learn. (CoRL)*, ser. Proc. Mach. Learn. Res., vol. 270, 2025, pp. 1949–1974.
- [11] J. Wang, M. Chen, N. Karaev, A. Vedaldi, C. Rupprecht, and D. Novotny, “VGGT: Visual geometry grounded transformer,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 5294–5306.
- [12] K. Black *et al.*, “ $\pi_{0.5}$: A vision-language-action model with open-world generalization,” in *Proc. Conf. Robot Learn. (CoRL)*, ser. Proc. Mach. Learn. Res., vol. 305, 2025, pp. 17–40.
- [13] Y. Du *et al.*, “Learning universal policies via text-guided video generation,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 36, 2023.
- [14] H. Wu *et al.*, “Unleashing large-scale video generative pre-training for visual robot manipulation,” in *Int. Conf. Learn. Represent. (ICLR)*, 2024.
- [15] M. Assran *et al.*, “Self-supervised learning from images with a joint-embedding predictive architecture,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 15 619–15 629.
- [16] A. Bardes *et al.*, “Revisiting feature prediction for learning visual representations from video,” *Trans. Mach. Learn. Res.*, 2024.
- [17] M. Assran *et al.*, “V-JEPA 2: Self-supervised video models enable understanding, prediction and planning,” arXiv:2506.09985, 2025.
- [18] G. Zhou, H. Pan, Y. LeCun, and L. Pinto, “DINO-WM: World models on pre-trained visual features enable zero-shot planning,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, ser. Proc. Mach. Learn. Res., vol. 267, 2025, pp. 79 115–79 135.
- [19] Y. Lin *et al.*, “JEPA-WAM: Learning vision-language-action policies with joint-embedding world modeling,” arXiv:2608.09381, 2026.
- [20] S. Yu *et al.*, “Representation alignment for generation: Training diffusion transformers is easier than you think,” in *Int. Conf. Learn. Represent. (ICLR)*, 2025.
- [21] S. Wang, V. Leroy, Y. Cabon, B. Chidlovskii, and J. Revaud, “DUST3R: Geometric 3D vision made easy,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 20 697–20 709.
- [22] V. Leroy, Y. Cabon, and J. Revaud, “Grounding image matching in 3D with MAST3R,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024.
- [23] Y. Li *et al.*, “WAM4D: Fast 4D world action model via spatial register tokens,” arXiv:2606.14048, 2026.
- [24] J. Zhang *et al.*, “Learning 4D geometric priors for inference-efficient world action models,” arXiv:2607.05468, 2026.
- [25] W. Peebles and S. Xie, “Scalable diffusion models with transformers,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 4195–4205.
- [26] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” in *Int. Conf. Learn. Represent. (ICLR)*, 2023.
- [27] Q. Bu *et al.*, “AgiBot World Colosseo: A large-scale manipulation platform for scalable and intelligent embodied systems,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2025.
- [28] J. Zheng *et al.*, “X-VLA: Soft-prompted transformer as scalable cross-embodiment vision-language-action model,” in *Int. Conf. Learn. Represent. (ICLR)*, 2026.
- [29] Wan-Video Team, “Wan2.2: Open and advanced large-scale video generative models,” Software release, 2025.
- [30] S. Bai *et al.*, “Qwen3-VL technical report,” arXiv:2511.21631, 2025.
- [31] T. Chen *et al.*, “RoboTwin 2.0: A scalable data generator and benchmark with strong domain randomization for robust bimanual robotic manipulation,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2026.